

Question number	Answer	Additional guidance	Mark
6(a)	• {presence / entry / invasion} of a pathogen inside {cells / tissues}	ACCEPT named pathogen e.g. bacteria / virus / viral genetic material / <i>Plasmodium</i> / sporozoites / merozoites named {cell / tissue} if correct IGNORE body	(1)

Question number	Answer	Mark
6(b)(i)	The only correct answer is B <i>A is incorrect because the rate = $229 \times 10^6 \div 4.09 \times 10^5 = 559.9 = 600$</i> <i>C is incorrect because the rate = $229 \times 10^6 \div 4.09 \times 10^5 = 559.9 = 600$</i> <i>D is incorrect because the rate = $229 \times 10^6 \div 4.09 \times 10^5 = 559.9 = 600$</i>	(1)

Question number	Answer	Mark
6(b)(ii)	The only correct answer is D <i>A is incorrect because $274\,000 : (409\,000 - 274\,000) = 274\,000 : 135\,000 = 2 : 1$</i> <i>B is incorrect because $274\,000 : (409\,000 - 274\,000) = 274\,000 : 135\,000 = 2 : 1$</i> <i>C is incorrect because $274\,000 : (409\,000 - 274\,000) = 274\,000 : 135\,000 = 2 : 1$</i>	(1)

Question number	Answer	Additional guidance	Mark
6(c)(i)	An explanation that includes three of the following points: • sporozoites more accessible to immune system (than other stages) (1) • because memory cells need to be generated (against the sporozoites) (1) • so that they will be destroyed by (secondary) immune response (1) • to prevent them {entering liver cells / damaging liver cells } (1)	ACCEPT converse idea for merozoites ACCEPT description of destruction by secondary immune response e.g. phagocytosis by macrophages BUT NOT IN THE CONTEXT OF THE PRIMARY RESPONSE DO NOT ACCEPT antibodies destroy ACCEPT before they can develop into merozoites	(3)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An explanation that includes three of the following points: - because {the immune response takes time to develop / phagocytes may not be able to engulf all the sporozoites} (1) - therefore a chemical needed to kill the sporozoites {in the vaccine / that get into the liver / so that they cannot produce merozoites} (1) - second chemical needed to kill the merozoites in case the first chemical does not kill all the sporozoites (1) - to avoid the risk of the person developing malaria (from the live sporozoites in the vaccine) (1)	ACCEPT the vaccine may be ineffective / the person may have a weakened immune system / the immune system may not destroy all the sporozoites ACCEPT weaken / reduce the number of IGNORE in case one chemical does not work	(3)

Question number	Answer	Additional guidance	Mark
6(c)(iii)	An explanation that includes three of the following points: • because the two strains of sporozoites had (some) same antigens (1) • and the vaccine stimulated {an immune response / memory cells / production of antibodies} (to these common antigens) (1) • because a secondary immune response took place (1) • and both types of sporozoites {were destroyed (by the immune system) / engulfed by macrophages} (1)	ACCEPT (surface) {molecules / proteins} ACCEPT (surface) {molecules / proteins}	(3)